Options
Title: Not titled yet
Output Language: Python
Generate Options: QT GUI

Variable
ID: samp_rate
Value: 1M

Variable
ID: sps
Value: 4

Variable
ID: excess_bw
Value: 350m

Variable
ID: rrc_taps
Value: firdes.root_raised_...

QT GUI Range
ID: noise_volt
Label: Channel: Noise Voltage
Default Value: 200m
Start: 0
Stop: 1
Step: 100m

Noise level of AAWGN added during transmission as well as antenna noise and LNA of receiver. in real case we dont need this?

QT GUI Range
ID: epsilon
Label: Channel: Timing Offset
Default Value: 1.0005
Start: 999m
Stop: 1.001
Step: 100u

Sets timing difference between transmitter and receiver due to variations in clocks

QT GUI Range
ID: freq_offset
Label: Channel:...ency Offset
Default Value: 100m
Start: -100m
Stop: 100m
Step: 1m

This accounts for differences in local oscillatiors as well as static Doppler shift

QT GUI Range
ID: delay
Label: Delay
Default Value: 52
Start: 0
Stop: 1k
Step: 1

This is used to align the transmitter and received bitstreams

Constellation Rect. Object
ID: bpsk
Symbol Map: 0, 1
Constellation Points: ...707m
Rotational Symmetry: 4
Real Sectors: 2
Imaginary Sectors: 2
Width Real Sectors: 1
Width Imaginary Sectors: 1

Symbol map: [0, 1] is Gray coded

Variable
ID: variable_0
Value: 0

File Source
File: ...BPSK_IMP/message.bin
Repeat: Yes
Add begin tag: ()
Offset: 0
Length: 0

Constellation Modulator
Constellation: <gnu...9941f0>
Differential Encoding: Yes
Samples/Symbol: 4
Excess BW: 350m

Throttle
Sample Rate: 1M
Limit: None

PlutoSDR Sink
IIO context URI: ip:...68.2.1
LO Frequency: 2.4G
Sample Rate: 1M
Buffer size: 32.768k
Cyclic: False
Attenuation TX1 (dB): 10
Filter Configuration: Auto
RF Bandwidth (Hz): 20M

Channel Model
Noise Voltage: 200m
Frequency Offset: 100m
Epsilon: 1.0005
Taps: 1
Seed: 0
Block Tag Propagation: No

Decimating FIR Filter
Decimation: 1
Taps: rrc_taps

QT GUI Frequency Sink
FFT Size: 1024
Center Frequency (Hz): 0
Bandwidth (Hz): 1M

QT GUI Time Sink
Number of Points: 200
Sample Rate: 1M
Autoscale: No

QT GUI Constellation Sink
Number of Points: 1.024k
Autoscale: No

FFT Root Raised Cosine Filter
Decimation: 1
Gain: 1
Sample Rate: 1M
Symbol Rate: 8k
Alpha: 350m
Num Taps: 44
Num. Threads: 1

QT GUI Frequency Sink
FFT Size: 1024
Center Frequency (Hz): 0
Bandwidth (Hz): 1M

QT GUI Time Sink
Number of Points: 200
Sample Rate: 1M
Autoscale: No

QT GUI Constellation Sink
Number of Points: 1.024k
Autoscale: No